

WEEK 4 – ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

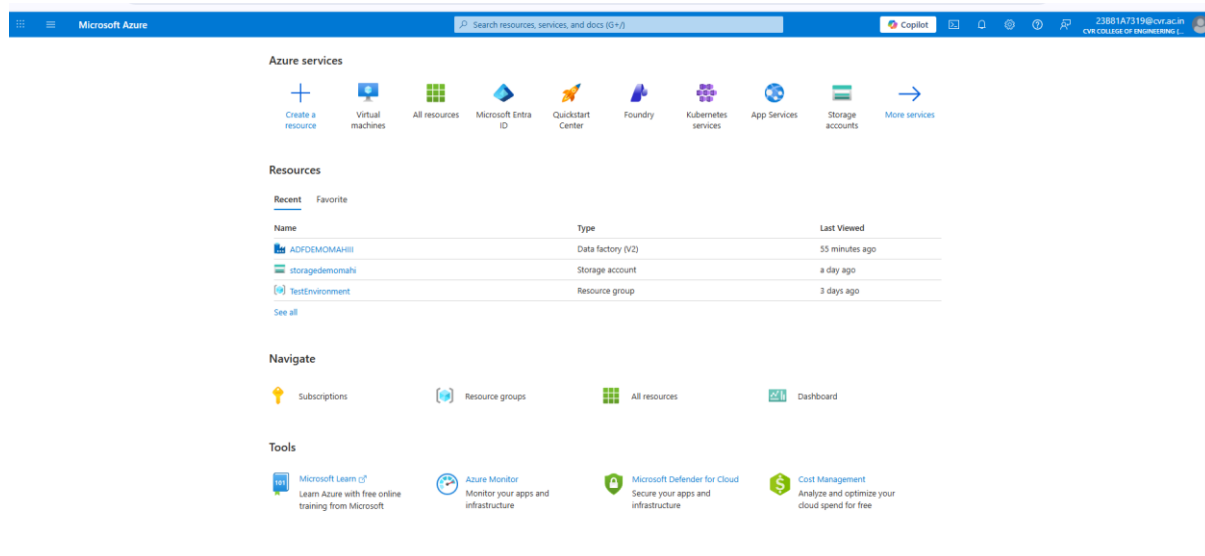
Assignment Tasks

Task 1:

- **Explore Azure Portal**

First, I logged in to the Azure Portal and explored the different services available in Azure. I became familiar with the portal interface, navigation menu, resource management options, and service categories such as Storage Accounts, Virtual Machines, Databases, and Azure Data Factory.

Screenshot 1: Azure Portal Interface



- **Create a Resource Group**

A Resource Group is a logical container that groups related Azure resources together. It helps in organizing, managing, monitoring, and controlling resources for a specific project.

Steps Performed:

1. Opened the Azure Portal.
2. Searched for Resource Groups in the search bar.
3. Clicked Create.
4. Selected the Azure subscription.
5. Entered the Resource Group name as CelebalAssignment.
6. Chosed the required region.

Screenshot 2: Creating the Resource Group "CelebalAssignment"

The screenshot shows the 'Create a resource group' page in the Azure Portal. The 'Basics' tab is selected. The page includes a description of a resource group and three input fields: 'Subscription' (set to 'Azure for Students'), 'Resource group name' (set to 'CelebalAssignment'), and 'Region' (set to '(Asia Pacific) Southeast Asia'). At the bottom, there are 'Previous', 'Next', and 'Review + create' buttons.

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure for Students

Resource group name * ⓘ CelebalAssignment

Region * ⓘ (Asia Pacific) Southeast Asia

Previous Next Review + create

7. Clicked Review + Create.

Screenshot 3: Resource Group Review Page

The screenshot shows the 'Review + create' tab of the 'Create a resource group' page. It displays a summary of the configuration: Subscription (Azure for Students), Resource group name (CelebalAssignment), and Region (Southeast Asia). There is an 'Automation Link' icon and a 'Tags' section. At the bottom, there are 'Previous', 'Next', and 'Create' buttons.

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

[Automation Link](#)

Basics

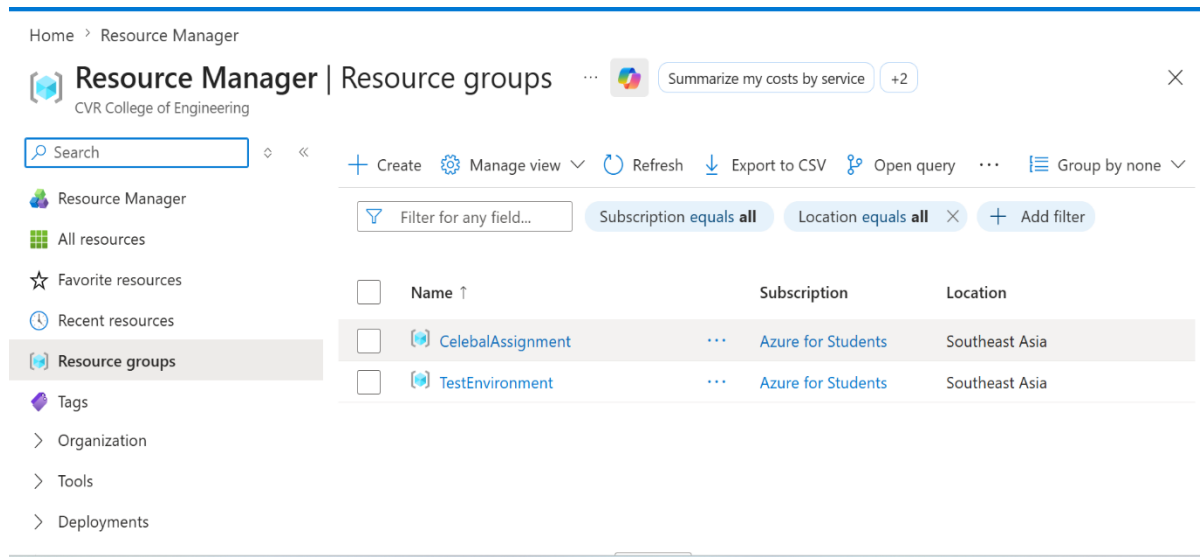
Subscription	Azure for Students
Resource group name	CelebalAssignment
Region	Southeast Asia

Tags

Previous Next Create

8. Validated the details and clicked Create.

Screenshot 4: Successfully Created Resource Group (CelebalAssignment)



Task 2: Storage Setup

• Create a Storage Account

I created an Azure Storage Account to store and manage data files. The Storage Account provides secure and scalable cloud storage services in Azure.

Steps Performed

1. Opened the Azure Portal.
2. Searched for Storage Accounts.
3. Clicked Create.
4. Selected the subscription and Resource Group (CelebalAssignment).
5. Entered the Storage Account name as celebaladfstorage.
6. Selected the required region.

Screenshot 5 : Creating the Storage Account " celebaladfstorage."

Home > Storage center | Blob Storage

Create a storage account

redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Instance details

Storage account name *

Region * [Deploy to an Azure Extended Zone](#)

Preferred storage type

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account)

7.Clicked Review + Create.

8.Clicked Create to deploy the Storage Account.

Screenshot 6: Successfully Created Storage Account " celebaladfstorage."

Home > celebaladfstorage_1781938176083 | Overview

celebaladfstorage

Storage account

[Improve storage account data protection](#) [Help me write an LCM rule for this storage account](#) [Create an LCM rule for my storage account](#)

[Upload](#) [Open in Explorer](#) [Delete](#) [Move](#) [Refresh](#) [Open in mobile](#) [CLI / PS](#) [Feedback](#)

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
 - Containers
 - Classic file shares
- Queues
- Tables
- Security + networking

[Add or remove favorites by pressing Ctrl+Shift+F](#)

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)
Container soft delete	Enabled (7 days)
Versioning	Disabled
Change feed	Disabled
NFS v3	Disabled
Allow cross-tenant replication	Disabled

Security

Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Public network access	Enabled
Public network access scope	Enable from all networks
Private endpoint connections	0

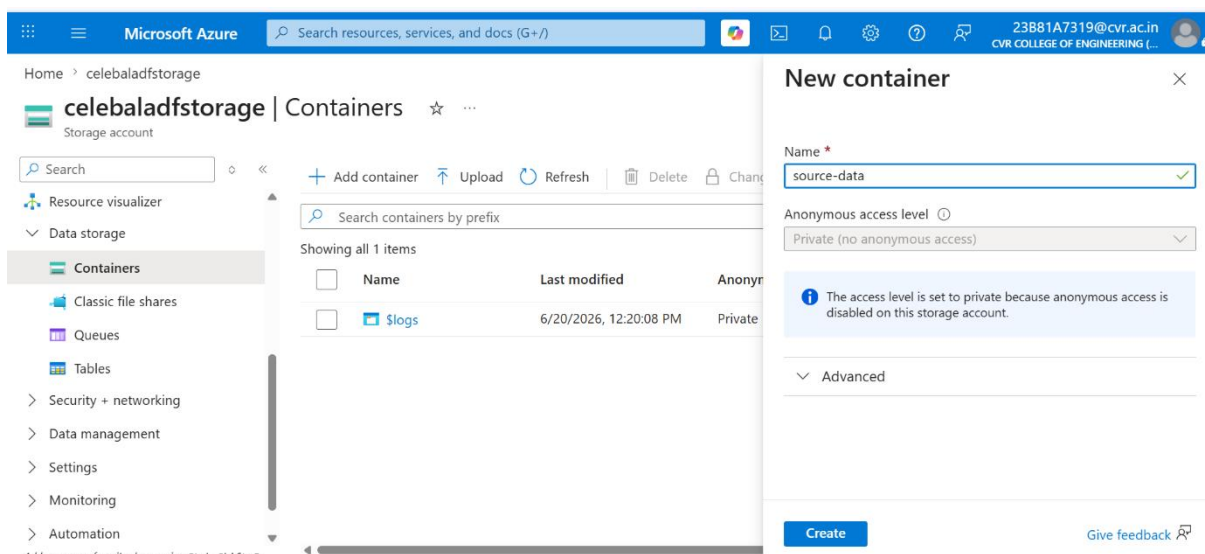
- **Create a Blob Container**

After creating the Storage Account celebaladfstorage, I created a Blob Container to store files and datasets.

Steps Performed

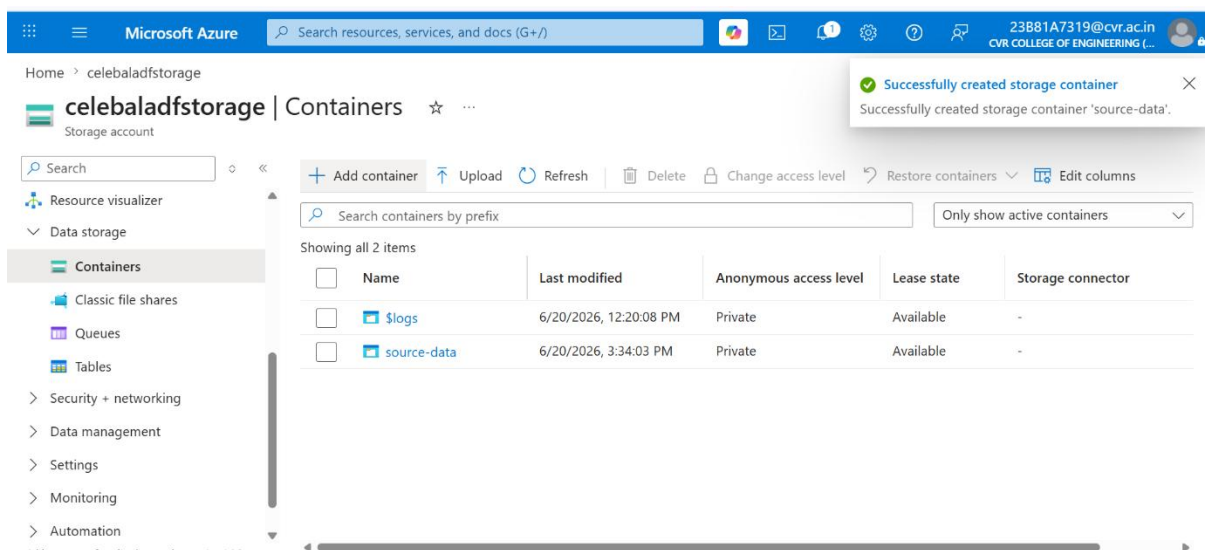
1. Opened the Storage Account celebaladfstorage.
2. Navigated to Data Storage → Containers.
3. Clicked + Container.
4. Entered a container name “source-data”.

Screenshot 7 : Creating the Blob Container " source-data."



5. Clicked Create.

Screenshot 8: Successfully Created Blob Container " source-data."



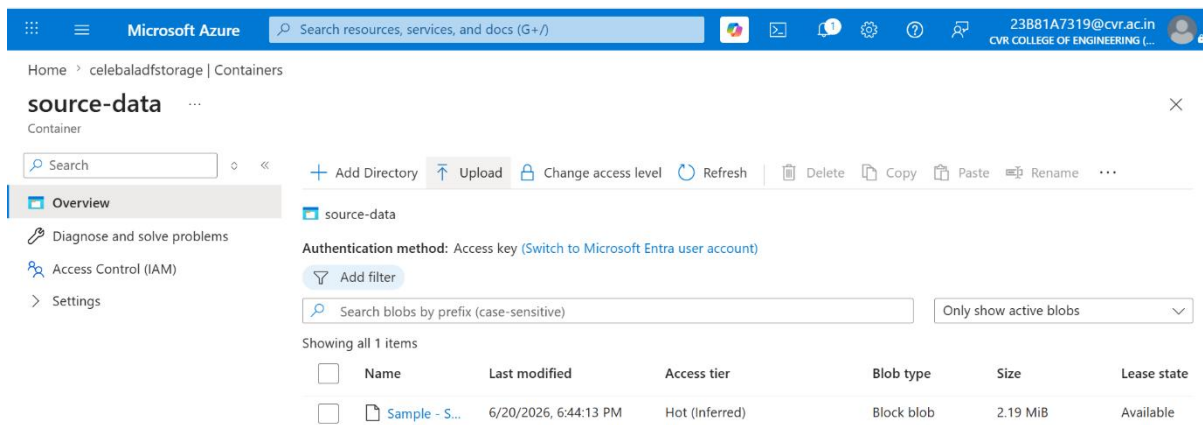
- **Upload a CSV file**

I uploaded a CSV file into the Blob Container

Steps Performed

1. Opened the source-data Blob Container.
2. Clicked the Upload button.
3. Selected the Sample – Superstore CSV file from the local system.
4. Clicked Upload to transfer the file to Azure Blob Storage.

Screenshot 8: Successfully Uploaded Sample-Superstore CSV File



Task 3: ADF Basics

- **Create Azure Data Factory**

I created an Azure Data Factory instance to build and manage data integration pipelines. Azure Data Factory is a cloud-based data integration service that enables data movement and transformation across different data sources.

Steps Performed

1. Opened the Azure Portal.
2. Searched for Azure Data Factory.
3. Clicked Create.
4. Selected the subscription and Resource Group (CelebalAssignment).
5. Entered the Data Factory name “**CelebalAssignment-AzureDataFactory**”
6. Selected the required region.
7. Clicked Review + Create.

Screenshot 9 : Creating the Azure Data Factory “CelebalAssignment-AzureDataFactory”

Microsoft Azure

Search resources, services, and docs (G+)

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CVR COLLEGE OF ENGINEERING

Home > Data Factory

Create Data Factory

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group *

CelebalAssignment

Create new

Instance details

Name *

CelebalAssignment-AzureDataFactory

Region *

(Asia Pacific) Southeast Asia

Version *

V2

We recommend you upgrade to Fabric Data Factory for a modern, unified data integration experience. Try Fabric instead!

Previous

Next

Review + create

Screenshot 10: Successfully Created Azure Data Factory “CelebalAssignment-AzureDataFactory”

Home > Microsoft.DataFactory-20260620190421 | Overview > CelebalAssignment

CelebalAssignment-AzureDataFactory

Data factory (V2)

How do I troubleshoot pipeline failures in this Data factory (V2)? +2

×

Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Monitoring

Automation

Help

Essentials

Resource group (move)
CelebalAssignment

Status
Succeeded

Location
Southeast Asia

Subscription (move)
Azure for Students

Subscription ID
d613a700-16cd-496d-968d-2a848e663381

Type
Data factory (V2)

Getting started
Quick start

JSON View

Azure Data Factory Studio

Add or remove favorites by pressing Ctrl+Shift+F

- **Explore ADF UI (Author, Monitor, Manage)**

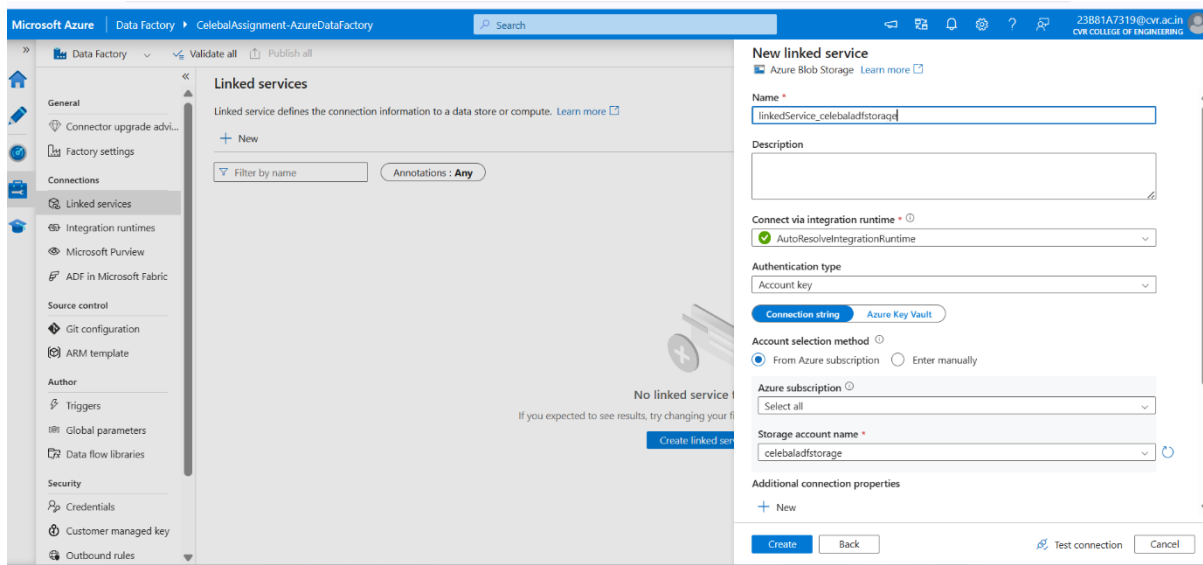
After creating Azure Data Factory, I launched ADF Studio and explored its main sections: Author, Monitor, and Manage.

- **Create Linked Service (Blob Storage)**

I created a Linked Service to establish a connection between Azure Data Factory and Azure Blob Storage. This connection allows ADF to access files stored in the Blob Container.

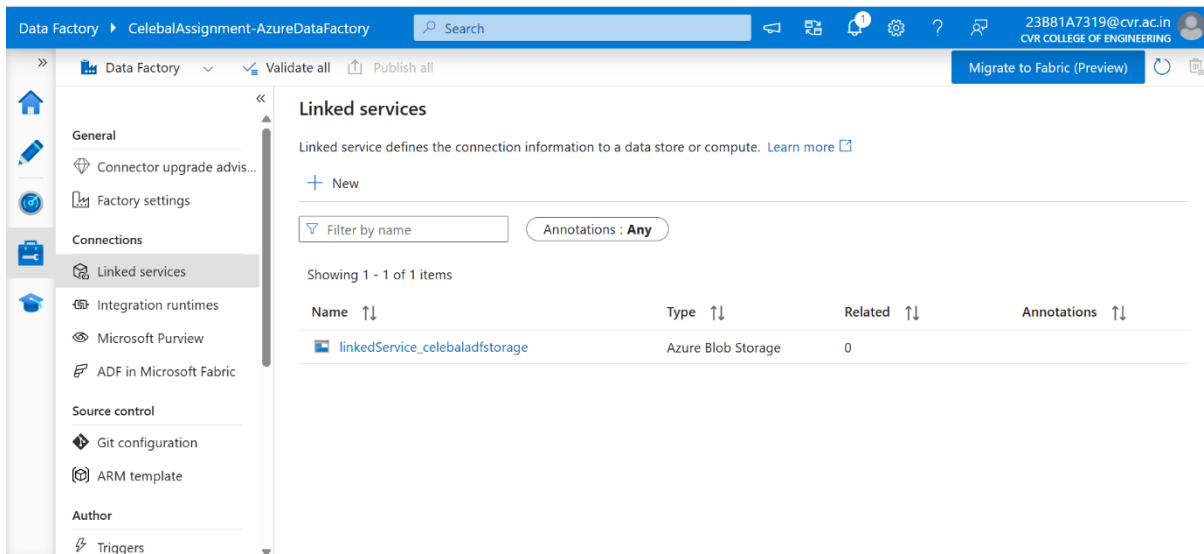
1. Opened ADF Studio.
2. Navigated to **Manage** → **Linked Services**.
3. Clicked + **New**.
4. Selected **Azure Blob Storage**.
5. Chose the authentication method.
6. Selected the Storage Account **celebaladfstorage**.

Screenshot 11 : Creating the Linked Service “linkedService_celebaladfstorage”



7. Clicked **Create**.

Screenshot 12: Successfully Created LinkService“linkedService_celebaladfstorage”



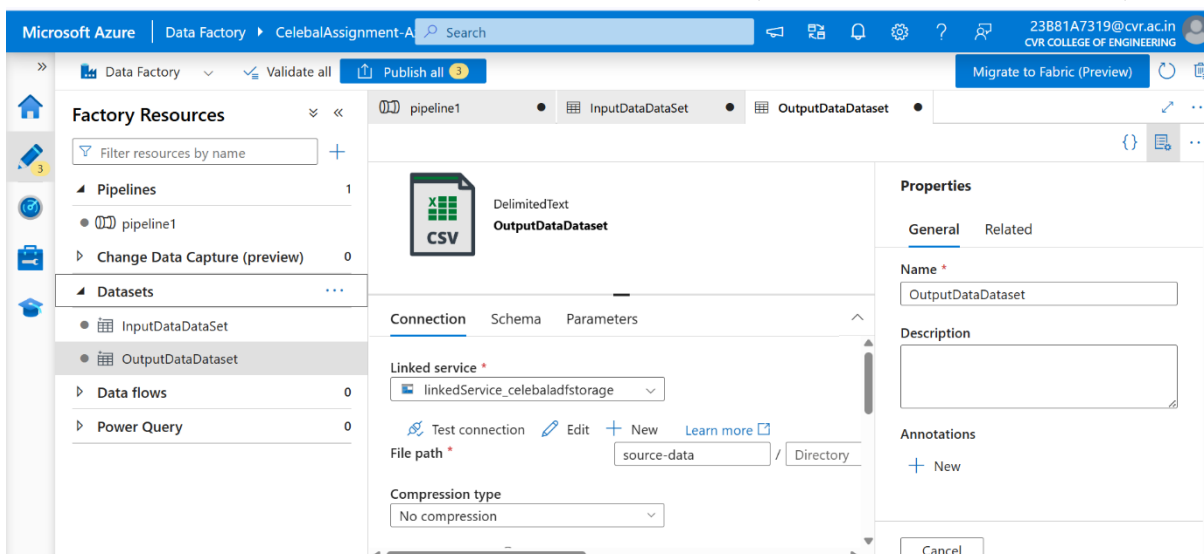
- **Create datasets (source + destination)**

I created datasets to define the source and destination locations for data processing in Azure Data Factory.

Steps Performed

1. Clicked + New Dataset.
2. Selected Azure Blob Storage.
3. Chose DelimitedText (CSV) as the file format.
4. Selected the same Blob Storage Linked Service.
5. Specified the destination container and output file path.
6. Saved the dataset.

Screenshot 13: Successfully Created datasets (source + destination)



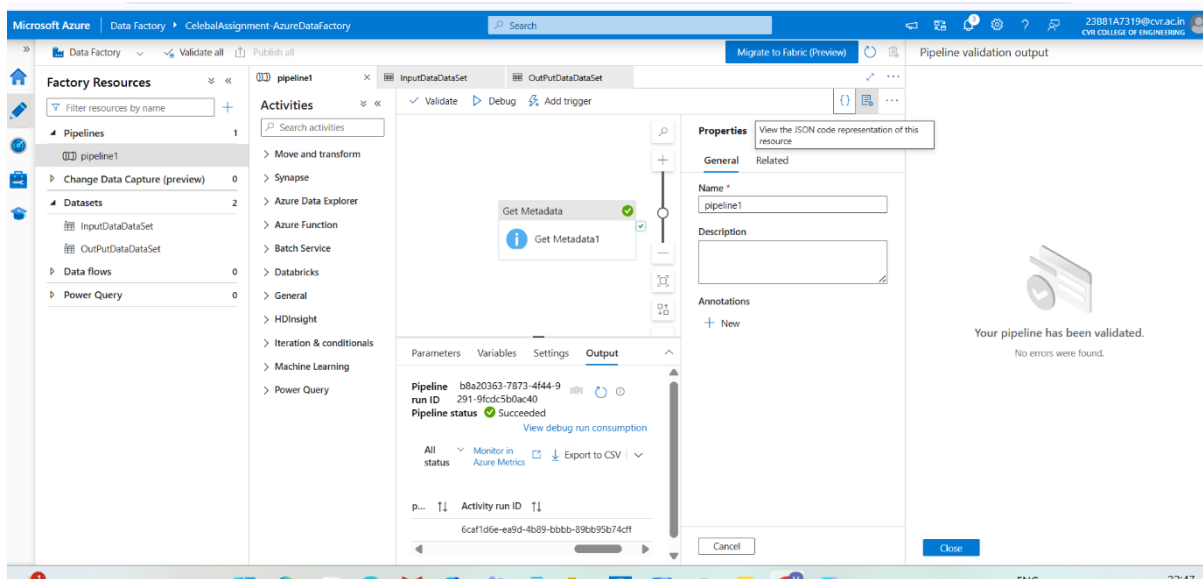
- **Use Get Metadata activity**

I used the Get Metadata activity in Azure Data Factory to retrieve information about the source CSV file stored in Azure Blob Storage. This activity helps verify file availability and obtain metadata such as file size, last modified date, and existence.

Steps Performed

1. Opened Azure Data Factory Studio.
2. Navigated to the Author section.
3. Created a new pipeline named pipeline1.
4. Dragged the Get Metadata activity from the Activities pane onto the pipeline canvas.
5. Selected the source dataset (InputDataDataSet).
6. In the Field List, selected the required metadata properties: Exists, Size, Last Modified, Item Name
7. Validated the pipeline configuration.

Screenshot 14: Successful Execution of Get Metadata Activity



Task 4: Pipeline Development

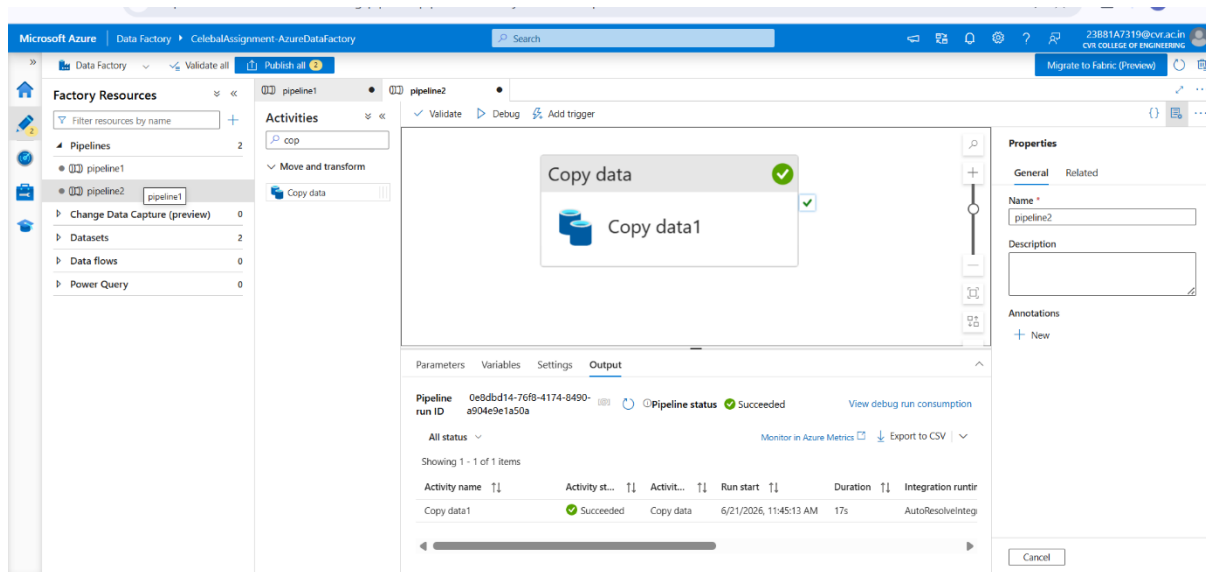
- **Create pipeline using Copy Data activity**

Steps Performed

1. Opened Azure Data Factory Studio.
2. Navigated to the Author section.
3. Created a new pipeline named pipeline2.

4. Dragged the Copy Data activity from the Move and Transform category onto the pipeline canvas.
5. Configured the Source dataset as InputDataDataSet.
6. Configured the Destination dataset as OutputDataDataset.
7. Validated the pipeline configuration.
8. Executed the pipeline using Debug.

Screenshot 15: Successful Execution of Copy Data Pipeline



• Configure source and destination

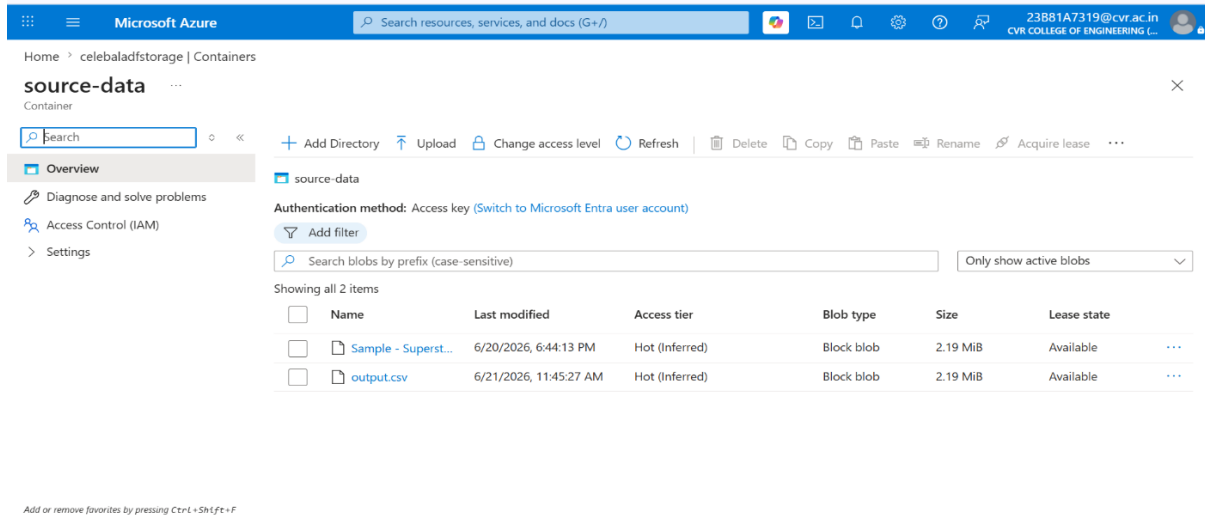
Source Configuration

- Dataset: InputDataDataSet
- File: Sample - Superstore.csv
- Container: source-data

Destination Configuration

- Dataset: OutputDataDataset
- File: output.csv
- Container: source-data

Screenshot 16: Output File Successfully Created in the source-data Container



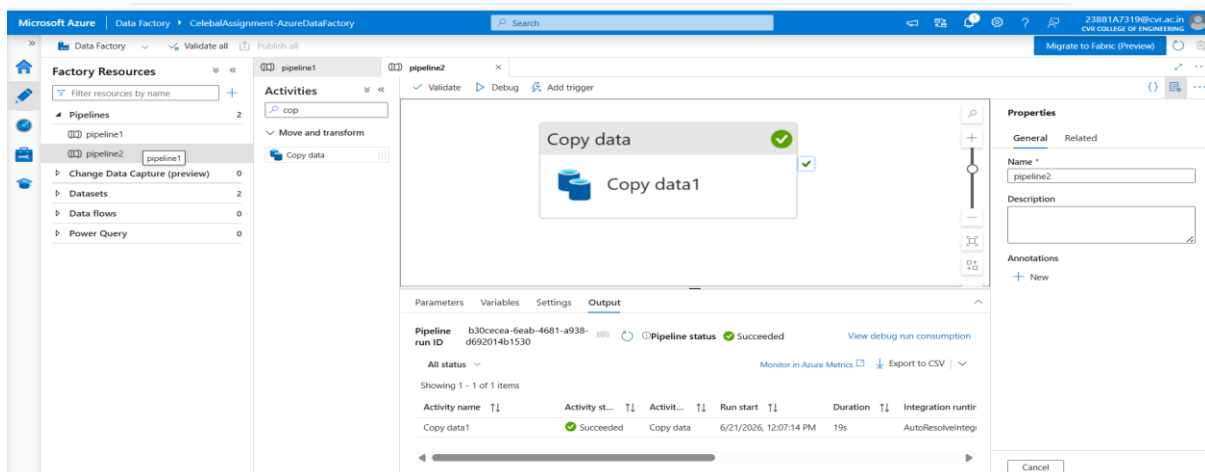
Task 5: Pipeline Execution

- Run pipeline using Debug/Trigger

Steps Performed

1. Opened Azure Data Factory Studio.
2. Navigated to the Author section.
3. Opened the pipeline2 containing the Copy Data activity.
4. Clicked Debug to execute the pipeline.
5. Monitored the pipeline execution.
6. Verified that the pipeline completed successfully with a Succeeded status.

Screenshot 17: Pipeline Execution Using Debug



Task 6: IAM Roles

- **Assign roles:**

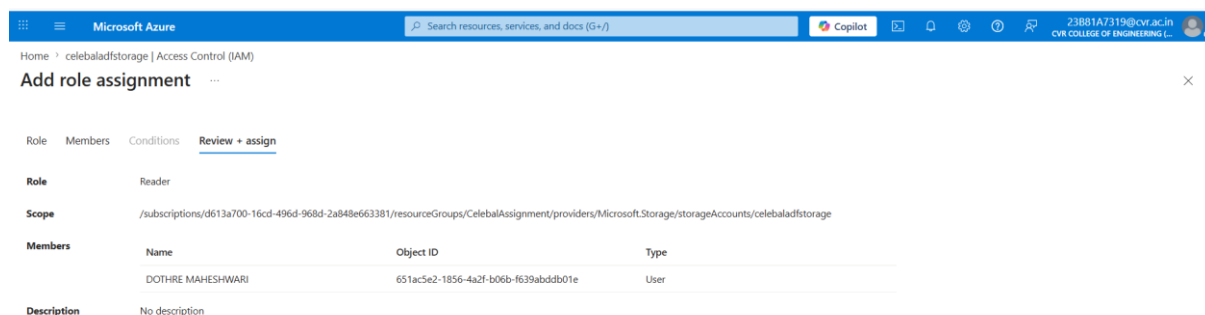
Reader

The Reader role provides read-only access to Azure resources. Users assigned this role can view resources and configurations but cannot make any changes.

Steps Performed:

1. Opened the Azure resource.
2. Navigated to Access Control (IAM).
3. Clicked Add ->Add Role Assignment.
4. Selected the Reader role.
5. Chose the required member.
6. Clicked Review + Assign.

Screenshot 18: Reader Role Assignment



Contributor

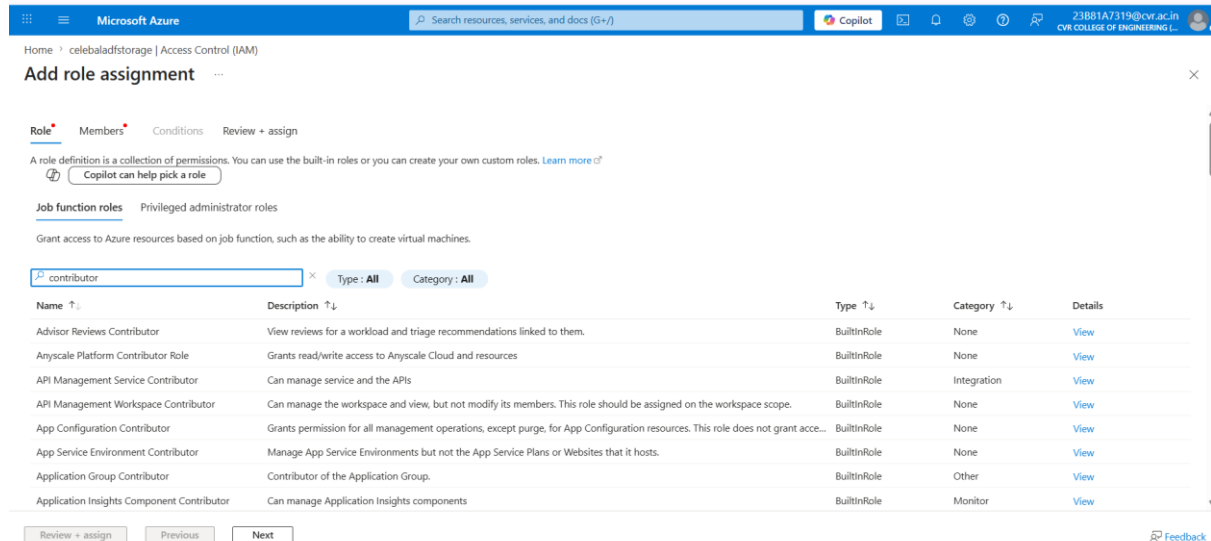
The Contributor role allows users to create, modify, and delete Azure resources but does not permit assigning roles to other users.

Steps Performed:

1. Opened Access Control (IAM).
2. Clicked Add ->Add Role Assignment.
3. Selected the Contributor role.

My Azure for Students subscription already provided the Owner role. Since the Owner role includes Contributor-level permissions, a separate Contributor role assignment was not available.

Screenshot 19: Contributor Role Not Available in Azure for Students Subscription



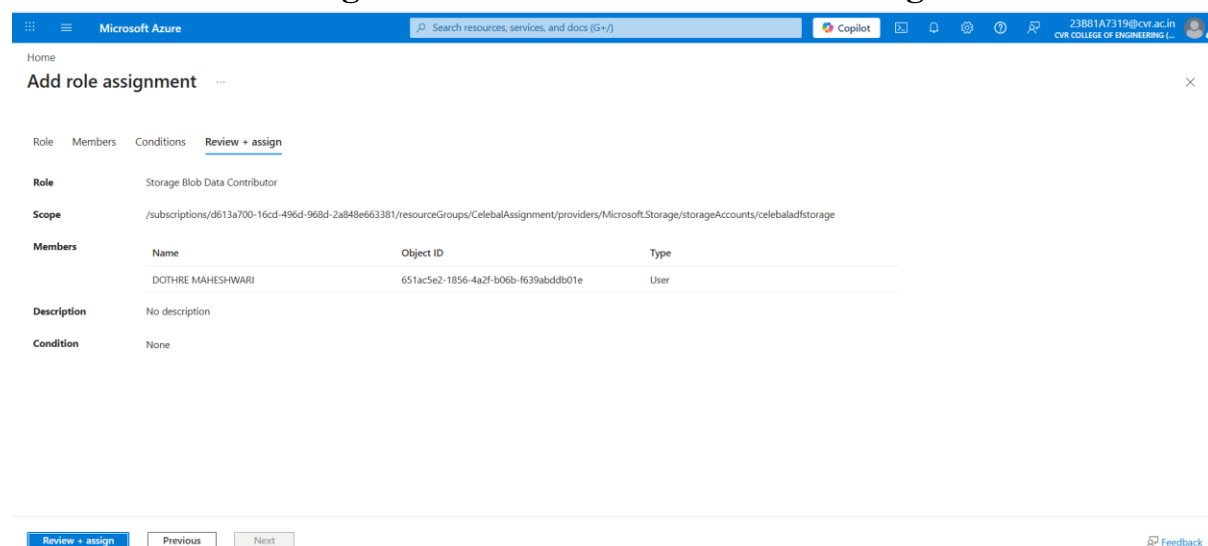
• Provide access to ADF for Storage

To enable Azure Data Factory to access files stored in Azure Blob Storage, the appropriate storage access role was assigned.

Steps Performed:

1. Opened the Storage Account (celebaladfstorage).
2. Navigated to Access Control (IAM).
3. Clicked Add -> Add Role Assignment.
4. Selected Storage Blob Data Contributor.
5. Chose the Azure Data Factory Managed Identity.
6. Clicked Review + Assign.

Screenshot 20: Storage Blob Data Contributor Role Assigned



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Description:

In this mini project, I created an end-to-end data pipeline using Azure Data Factory. The pipeline reads the Sample - Superstore.csv file from Azure Blob Storage, validates the file using the Get Metadata activity, and copies the data to a new destination file using the Copy Data activity.

Components Used

- Azure Blob Storage(celebaladfstorage)
- Linked Service(linkedService_ celebaladfstorage)
- Source Dataset (InputDataDataSet)
- Destination Dataset (OutputDataDataset)

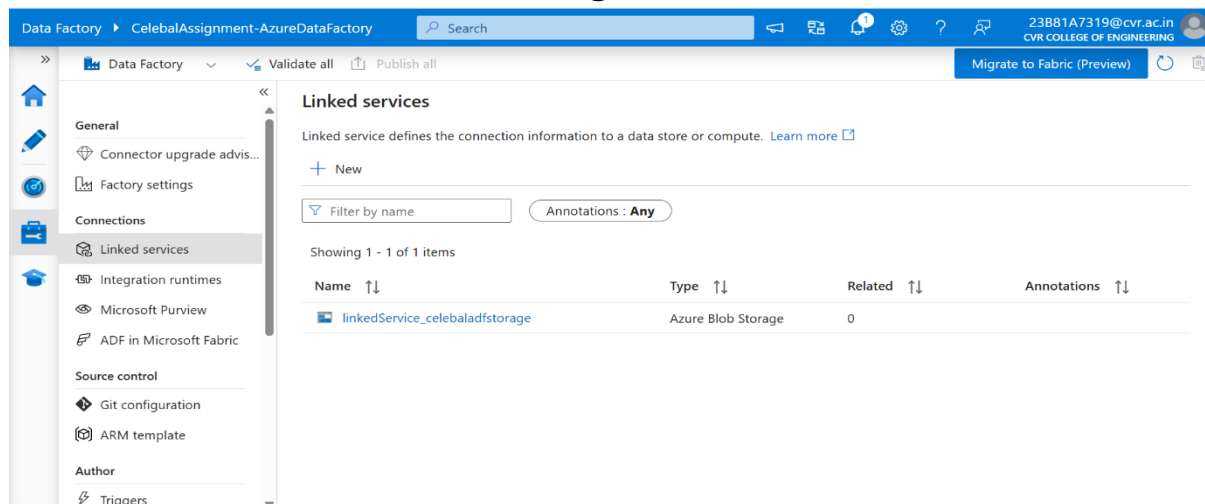
- Get Metadata Activity
- Copy Data Activity
- Azure Data Factory Pipeline

Implementation Steps

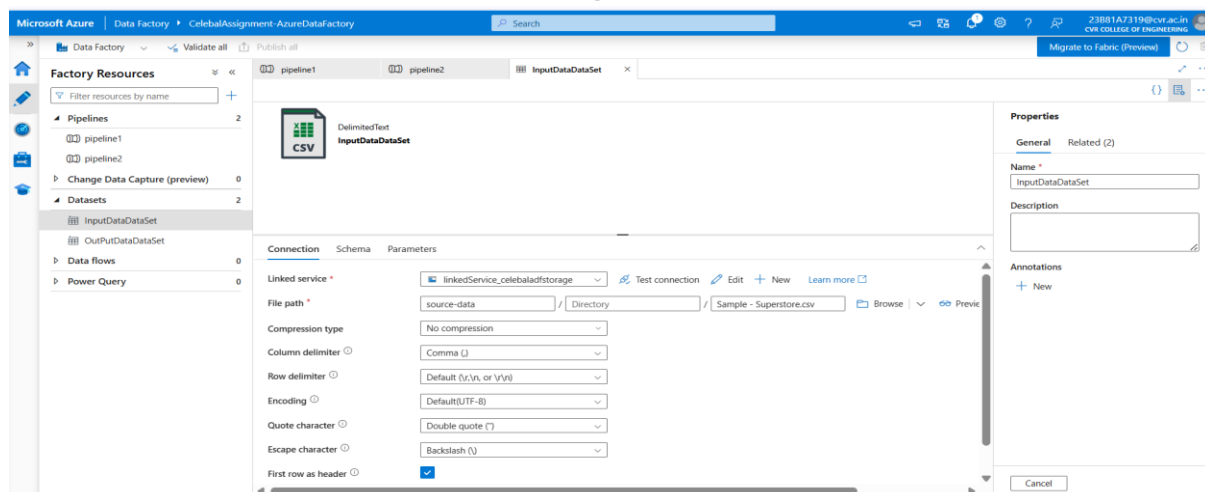
1. Uploaded the Sample - Superstore.csv file to the source-data Blob Container.
2. Created a Linked Service to connect Azure Data Factory with Azure Blob Storage.
3. Created source and destination datasets.
4. Configured the Get Metadata activity to validate the source file.
5. Added the Copy Data activity to copy data from the source file to the destination file.
6. Executed the pipeline using the Debug option.
7. Verified successful pipeline execution

Output

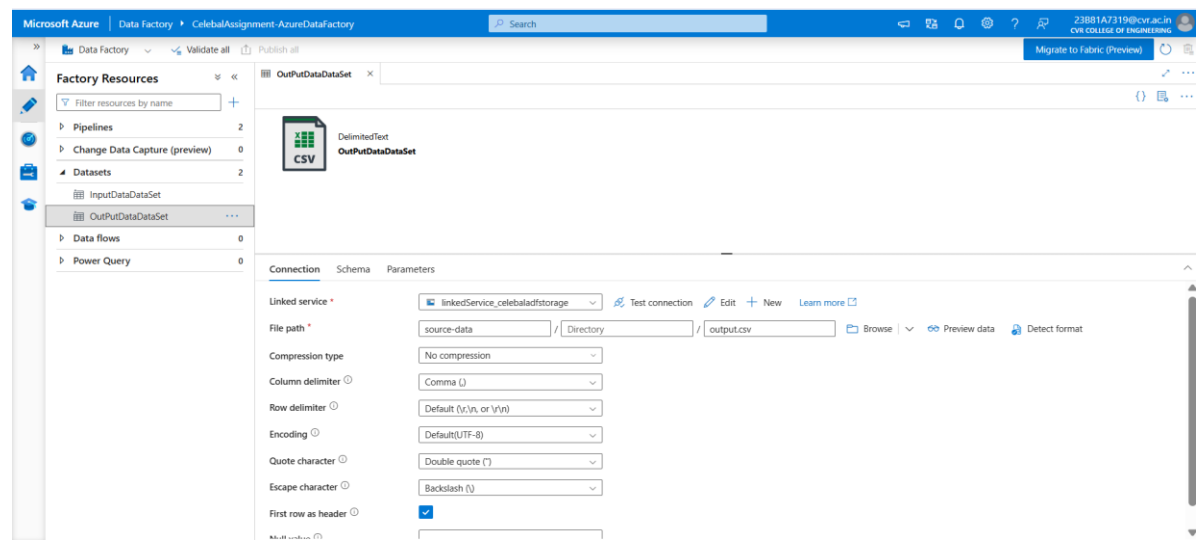
Screenshot 18: Linked Service Configuration



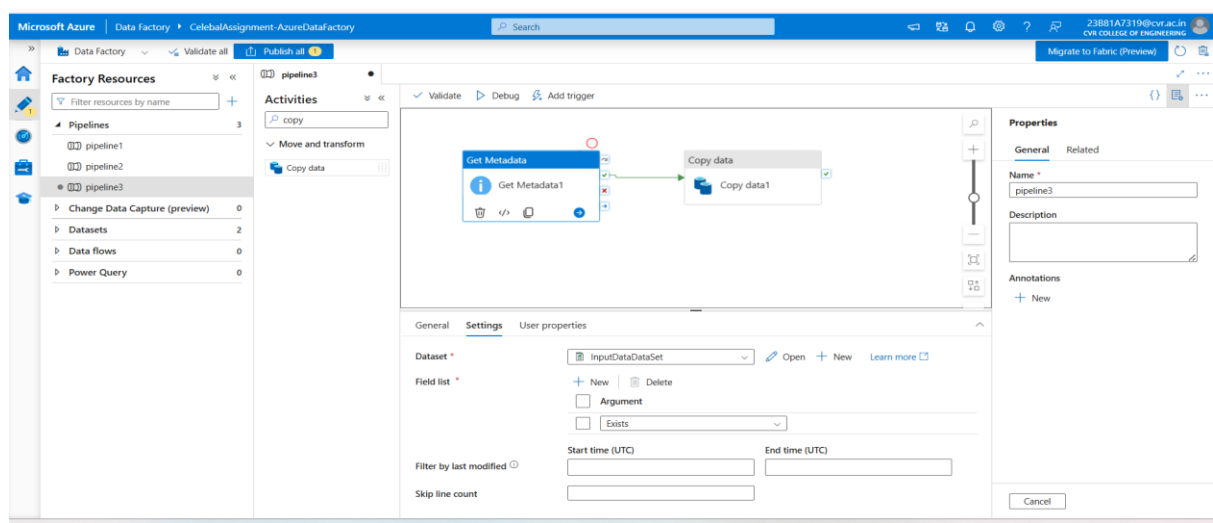
Screenshot 19: Source Dataset Configuration



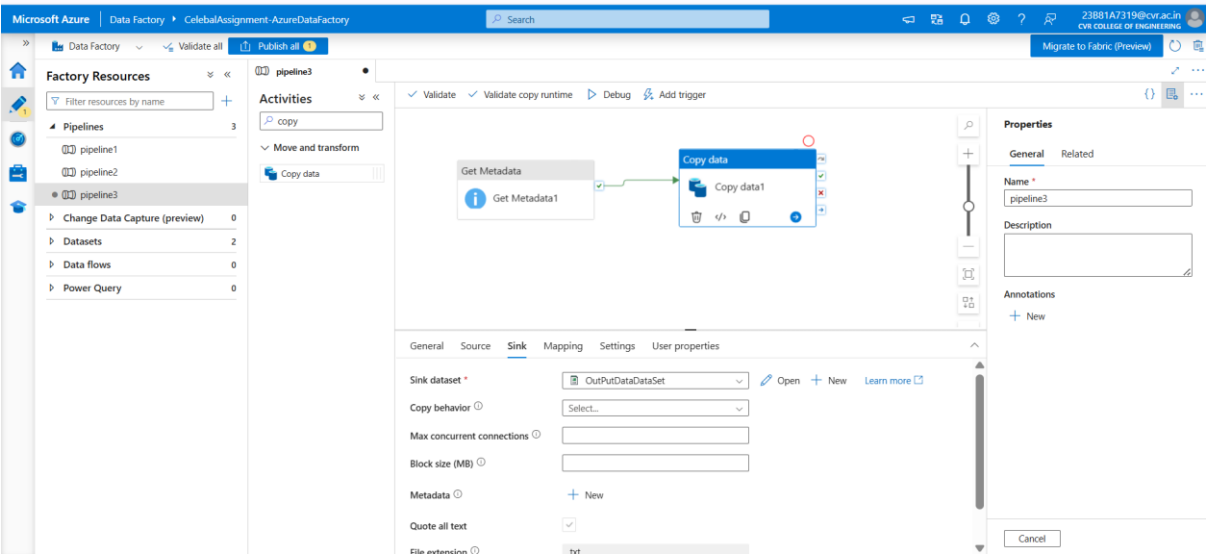
Screenshot 20: Destination Dataset Configuration



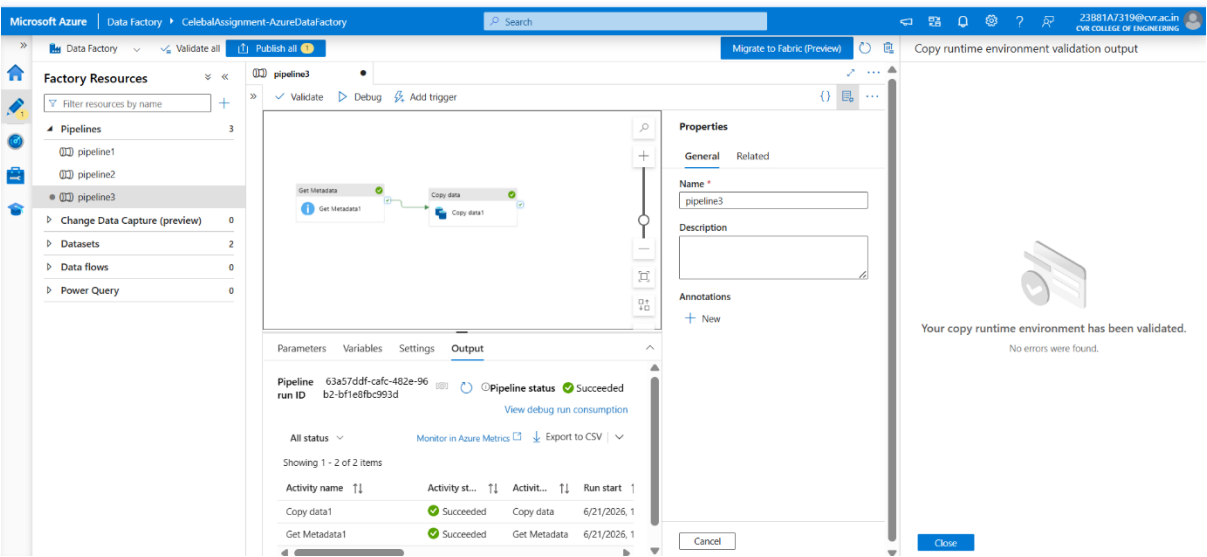
Screenshot 21: Get Metadata Activity Configuration



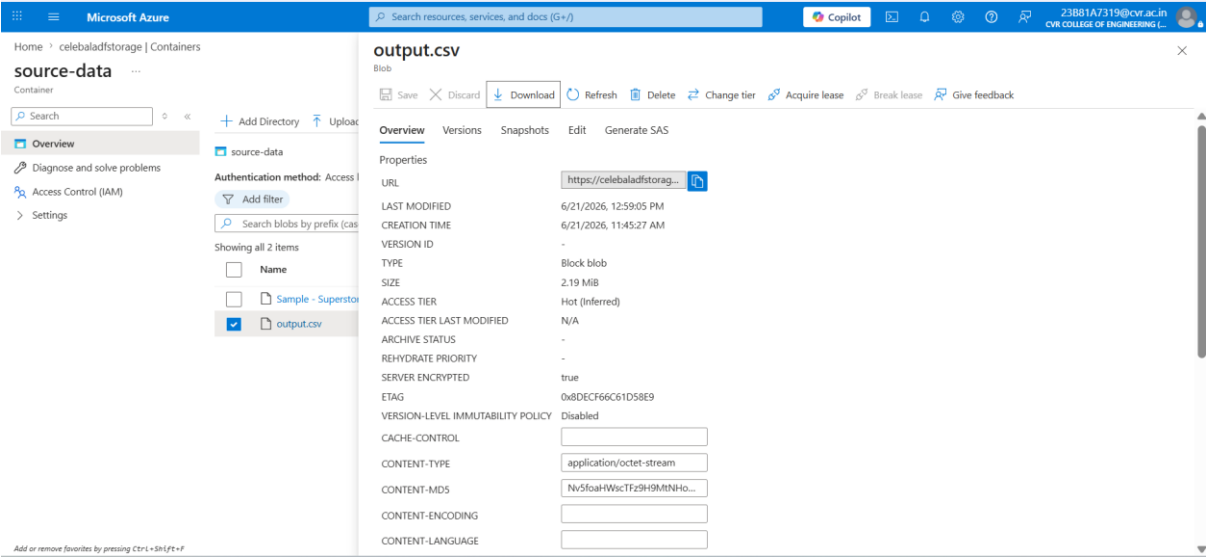
Screenshot 22: Copy Data Pipeline Design



Screenshot 23: Pipeline Execution Status Succeeded



Screenshot 24: Output File Successfully Created in Blob Storage



Result

The pipeline executed successfully. The source CSV file was validated using the Get Metadata activity, and the data was copied successfully to the destination file in Azure Blob Storage. The successful execution confirmed that the Linked Service, Datasets, and Pipeline were configured correctly.